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React Suspense

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React Suspense lets you display an alternative HTML while waiting for code or data to load.

The alternative HTML can be a component, text, or any valid content.

What is Suspense?

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Suspense is a React feature that lets your components display an alternative HTML while waiting for code or data to load.

The most common use cases are:

- Data fetching with suspense-enabled frameworks
- Loading components dynamically with `React.lazy()`

Using Suspense

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Example

Get your own React.js Server

The `Fruits` component takes two seconds to load, so we wrap it in a `Suspense` component to display a loading message while it is loading.

```
import { createRoot } from 'react-dom/client';
import { Suspense } from 'react';
import Fruits from './Fruits';

function App() {
  return (
    <div>
      <Suspense fallback={<div>Loading...</div>}>
        <Fruits />
      </Suspense>
    </div>
  );
}

createRoot(document.getElementById('root')).render(
  <App />
);
```

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Using Suspense with `lazy` Loading

Another common use of the `Suspense` component is when importing components with `lazy` loading:

In the example above we had to fake a delay of two seconds to see the loading message. A task like displaying three fruits from an array would be too fast to see the loading message at all.

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seconds delay:

Example

This example is too fast to see the loading message:

```
import { createRoot } from 'react-dom/client';
import { Suspense } from 'react';
import Cars from './Cars';

function App() {
  return (
    <div>
      <Suspense fallback={<div>Loading...</div>}>
        <Cars />
      </Suspense>
    </div>
  );
}

createRoot(document.getElementById('root')).render(
  <App />
);
```

Run Example »

Now let us create an example WITH using `lazy` loading:

Example

Same example as above, but using `lazy` loading:

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```
const Cars = lazy(() => import('./Cars'));

function App() {
  return (
    <div>
      <Suspense fallback={<div>Loading...</div>}>
        <Cars />
      </Suspense>
    </div>
  );
}

createRoot(document.getElementById('root')).render(
  <App />
);
```

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Example Explained

- `lazy()` lets you load a component dynamically
- `Suspense` shows a fallback while the component loads

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Multiple Components

One Suspense component can wrap multiple lazy components:

Example

```
import { createRoot } from 'react-dom/client';
import { Suspense, lazy } from 'react';

const Header = lazy(() => import('./Header'));
const Content = lazy(() => import('./Content'));
const Sidebar = lazy(() => import('./Sidebar'));

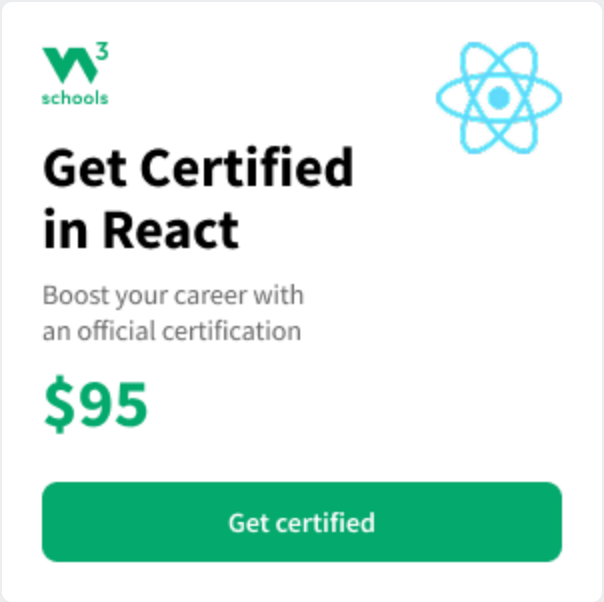
function App() {
  return (
    <div>
      <Suspense fallback={<div>Loading...</div>}>
        <Header />
        <div style={{ display: 'flex' }}>
          <Sidebar />
          <Content />
        </div>
      </Suspense>
    </div>
  );
}

createRoot(document.getElementById('root')).render(
  <App />
);
```

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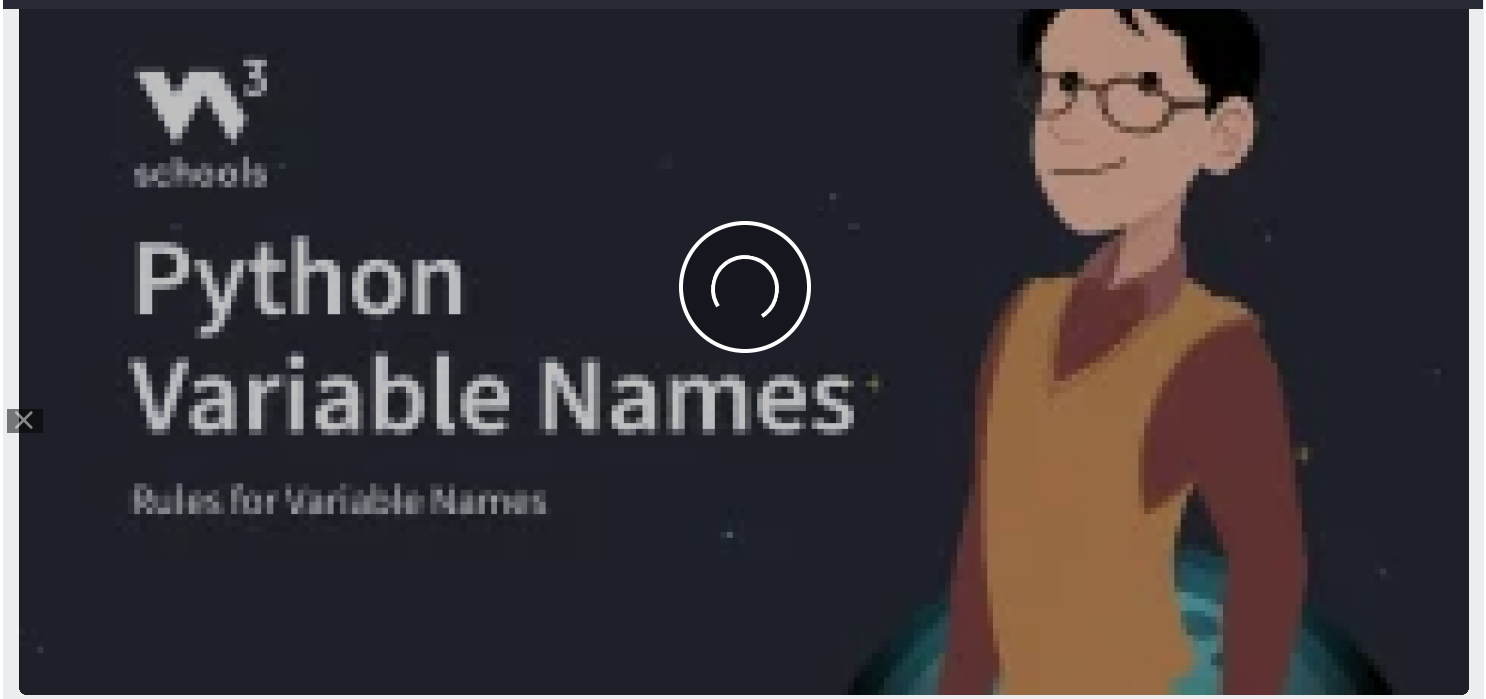
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